HARVARD CENTER FOR INTERNATIONAL DEVELOPMENT

Competing in Times of Cognitive Capital

Kansas At A Crossroads

Mark Esposito, PhD
Harvard Kennedy School
Berkman Klein Center for Internet & Society

mesposito@cyber.harvard.edu

Building a thriving world for all

Remobilizing in Globalization

Digital Forces Are Reimagining Society's Largest Touch Points

Foundational

- Remanufacturing the Economy
- Trade Resilience
- Healthcare Burden

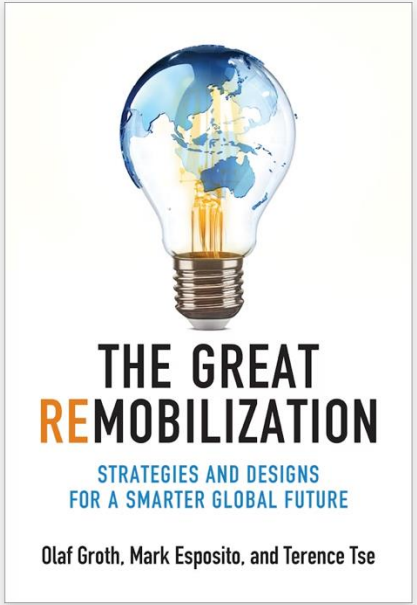
Emerging

- Health + Tech Integration
- AgriTech & Food Systems
- Data Centers as Factories

Future-Defining

- Quantum Economy
- Cognitive Capital
- Geostrategy of Data

Source: CIGI, The Manufacturer, McKinsey, SRI International, World Economic Forum



The Great Remobilization

Strategies and Designs for a Smarter Global Future

By Olaf Groth, Mark Esposito and Terence Tse

Hardcover +

eBook +

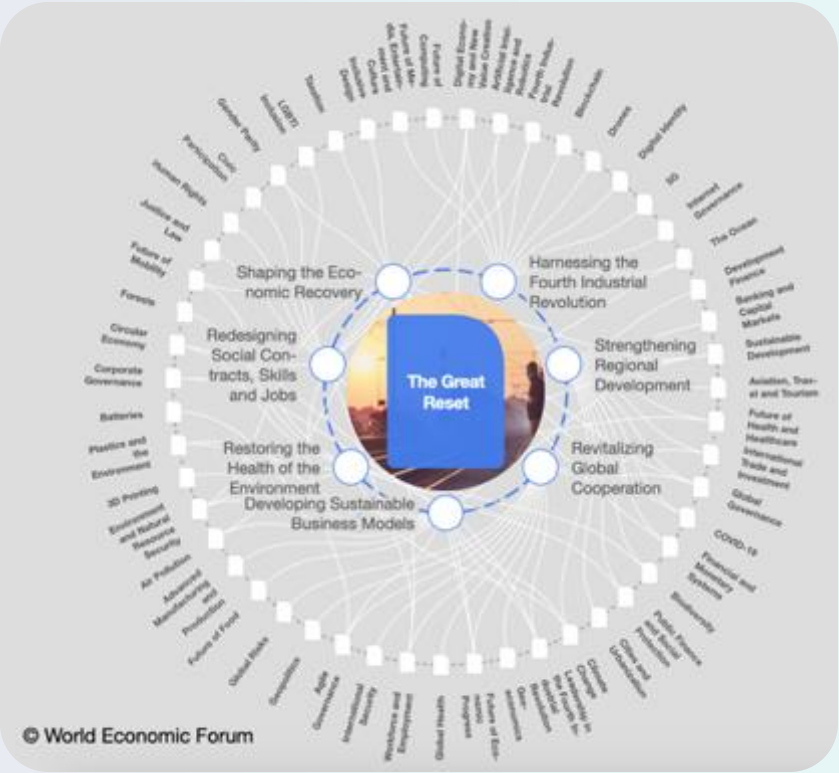
- 2024 Axiom Business Book Awards - Silver Medal in International Business / Globalization
- 2024 PROSE Award Finalist: Business, Finance, and Management
- Shortlisted for the Thinkers50 Strategy Award.

The Great Reset Initiative

Overcoming the pandemic and reimagine our economy

"The pandemic represents a rare but narrow window of opportunity to reflect, reimagine, and reset our world"

Klaus Schwab
Founder and executive chairman, world economic Forum



Geopolitics Has Become a Technology Game

Geopolitics has shifted from the battlefield to the cloud, where digital factories, satellites, and algorithms decide which nations rise and which fall.

Cognitive Capital

Peak of AI

Digital Capital

Results of COVID

Industrial Capital

Trump Admin on US Manufacturing

2000 2010 2020 2030

1) Owning the Invisible Factories

- Data centers, cloud hubs, and quantum labs are today's factories
- Nations compete for sovereignty through control of **data flows**

Google's new plan to keep its data centers from stressing the grid

AI data centers are set to swamp U.S. power grids. Google's new utility deals aim to flip the script — shifting workloads to ease strain and speed its access to energy.

By Jeff St. John
28 August 2025

2) Weaponizing Information

- Satellites, AI, and geospatial intelligence redefine security
- Control of **algorithms = control of outcomes** in geopolitics

Ukraine war: Satellite images reveal Russian defences before major assault

21 May 2023

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Daniele Palumbo & Erwan Rivault BBC Verify

3) Programming the Economy

- Digital currencies and algorithmic pricing reshape exchange
- Whoever writes the code, writes the rules of globalization

Will AI lead to abundance? Exploring cost reductions from streaming to tokenized technologies

Mfon Akpan

Show more

Source: Google, BBC, Medium, U.S. Government (Trump Administration), COVID-19 Global Health Event

The economy needs to adapt to AI as a factor of production

The future economy will be built on algorithms, not assembly lines.

Production Shifts

- ”America First” Industrial Narrative
- Policy Realignment Toward Protectionism + Industrial Sovereignty

Monetary Transformation

- **CBDCs = Sovereignty Tools** (money coded as independence)
- Dynamic Pricing = +$0.5T Value by 2030

Data as Capital

- AI converts raw data into capital
- AI attention markets replace price signals

AI redefines the epistemology of value: shifting production from labor to learning systems.

Source: U.S. Government (“America First” industrial policy), Central Banks (CBDCs), Global Financial Institutions (Dynamic Pricing projections), Artificial Intelligence Research Community

What Happens When Technology Outpaces Control?

When algorithms become infrastructure, mistakes become systemic.

AXIOS

23 hours ago - Business

Race with China divides AI leaders, policymakers

Scott Rosenberg

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Reuters

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US launches effort to speed power grid projects for AI

By Timothy Gardner

September 18, 2025 3:20 PM EDT · Updated 12 hours ago

Aa

Global inequality + Add to myFT

AI risks widening global wealth gap, WTO warns

Poorer countries need to be able to share benefits of artificial intelligence, report says

Albert Fox • Follow

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Geopolitical Tensions and Tech Reallocation: China's AI Chip Push and the Rise of Domestic Contenders

Source: Axios, Associated Press, Reuters, Financial Times, World Trade Organization, Albert Fox

Kansas: An Economic Aurora

Kansas is redefining the geography of innovation in America’s economy.

An economic aurora describes the sudden illumination of a region’s latent capacities: in Kansas, the convergence of data infrastructure, advanced manufacturing, and biomanufacturing has transformed the state from a quiet corridor into a visible frontier of global competitiveness.

The Hidden Engine That Could Reinvent How We Produce by 2075

Kansas's future depends not only on agriculture and manufacturing, but on water sovereignty, bio-manufacturing, and data infrastructure – factors that could redefine production by 2075.

Water as Wealth

Energy Transition Hub

Bio-Industrial Base

Data Center Geography

AgriTech + Robotics

Supply Chain Redesign

Source: U.S. Geological Survey (Ogallala Aquifer), U.S. Department of Energy (Energy Transition), Kansas State Government (Bio-Industrial Initiatives), Data Center Industry Reports, Agricultural Technology Companies, Supply Chain and Logistics Reports

Why Kansas Matters Beyond Kansas

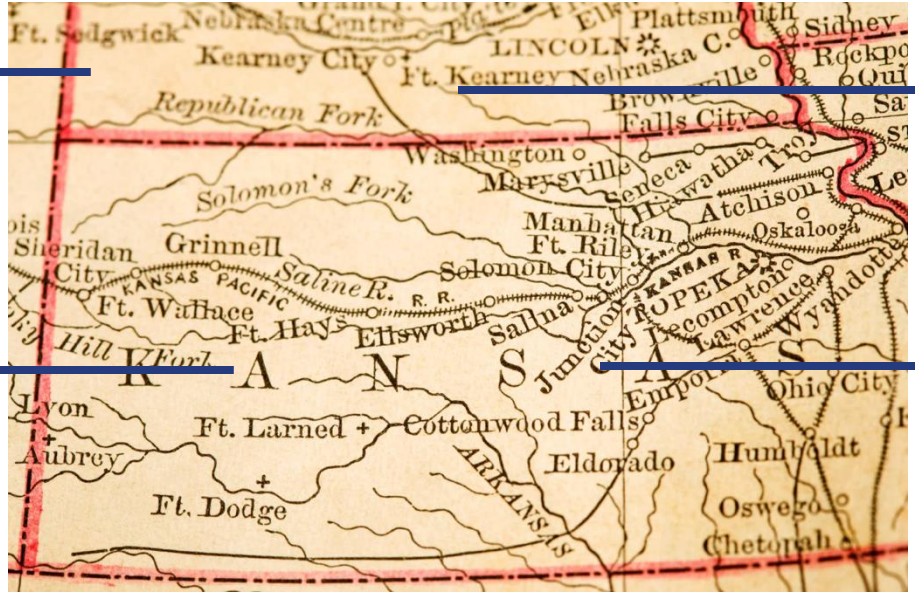
2075 Will Belong to Regions That Anchor the Digital in the Physical.

Renewable Heartland

- Top-tier wind farms, solar corridors, hydrogen + grid storage pilots

Aerospace + AI Skies

- Spirit AeroSystems, legacy Boeing facilities, drone R&D, automation in aviation



Digital Backbone

- Google + Meta data centers, Port KC tech projects, interstate + fiber convergence

Bio-Industrial Frontier

- Kansas State University → ag-biotech/cancer research, biomanufacturing hub, food security innovation

Source: Spirit AeroSystems, Boeing, Google, Meta, Port KC, Kansas State University, U.S. Renewable Energy Reports

Kansas Is Uniquely Positioned to Excel in the Global Economy

Other U.S. regions excel in single domains, but Kansas fuses water, energy, food-tech, and bio-industry into a future-ready model.

California
Innovation but Fragile

- Tech + venture capital concentration
- Water scarcity + climate vulnerability
- Dependence on imports for food & energy

Kansas
Integration as Sovereignty

- Water = Strategic Sovereignty
- Bio-Industrial Proteins, Not Just Crops
- Digital Roots in Rural Soil

Texas
Energy Giant but Exposed

- Oil & gas dominance (transition risk)
- Strong wind power, but grid instability
- Urban growth stresses infrastructure

Source: California State Reports (Water & Climate), Texas Energy and Infrastructure Reports, Kansas State Research Institutions (Water, Bio-Industry, Digital Infrastructure)

Why Kansas Becomes the Laboratory

Small enough to experiment, big enough to matter.

Agri-Tech Pilots → Scalable Food Security

Kansas fields as controlled test zones for robotics, AI irrigation, and resilient crops

Data Centers → Rural Digital Backbone

Hyperscale projects stress-test grid, water, and sovereignty models

Biomanufacturing Hubs → New Industrial Base

Universities + private sector shaping proteins, bio-materials, and health inputs

Energy Corridors → Transition Stress Test

Wind, solar, and hydrogen scaled in real-world, not theoretical, conditions

Kansas is the laboratory where water, energy, food, and data converge to prototype the sovereignty frameworks of 2075.

Source: Agricultural Technology Companies, Hyperscale Data Center Reports, Kansas Universities and Biomanufacturing Research, U.S. Renewable Energy Projects (Wind, Solar, Hydrogen)

Kansas as the Stress Test for Global Resilience

Climate Volatility Lab

Grid & Energy Stability

Food-Tech Resilience

Digital Sovereignty Proof

Supply Chain Fragility

Population Shifts Testbed

Kansas operates as a resilience crucible: its outcomes will preview which systems endure into 2075.

Source: Climate Science Reports, Energy Grid Reliability Studies, Food Technology Research, Digital Governance and Sovereignty Studies, COVID-19 Public Health Data, Supply Chain Resilience Analyses, Demographic and Urbanization Reports

Signals of the Aurora

The future economy isn't abstract: Kansas is already showing early flashes of 2075 sovereignty systems.

Kansas Actions / Signals

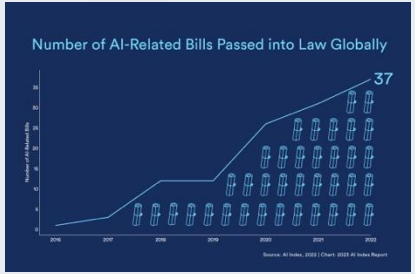
Panasonic breaks ground on $4B EV battery plant in Kansas: largest private investment in state history

Agri-data platforms in Kansas fields tested for AI-driven irrigation and yield optimization

Wichita aerospace labs integrate drones + AI for next-gen aviation automation

What These Signals Point To by 2075

- AI-Optimized EV Corridors anchoring Midwest supply chains
- Agri-Data as Sovereign Capital in a digitized food economy
- Aerospace + AI Automation exporting resilience worldwide



7x

Increase in AI-related laws passed globally since 2016.

Key Takeaway: Kansas's testbeds in automation will only succeed if paired with similar leaps in governance capacity.

Source: Panasonic, Kansas Agricultural Data Initiatives, Wichita Aerospace Research Labs, Global AI Policy Reports (OECD/UNESCO/AI Index)

Rethinking Growth Through the Digital Commons

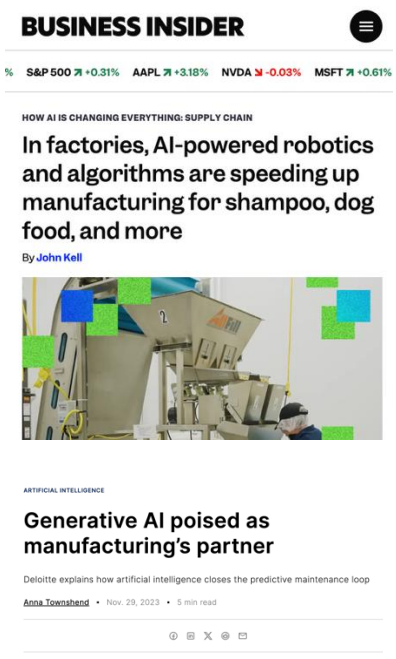
AI Age – A democratic promise	Mechanism	Challenges
The Commons as Capital	Shared water, energy, and data infrastructures	Few firms dominate compute, energy grids, and data flows
AI Governance in Practice	Open digital governance models (AI, blockchain)	Open digital governance models (AI, blockchain)
From Local to Global Model	Distributed innovation beyond Silicon Valley	Distributed innovation beyond Silicon Valley

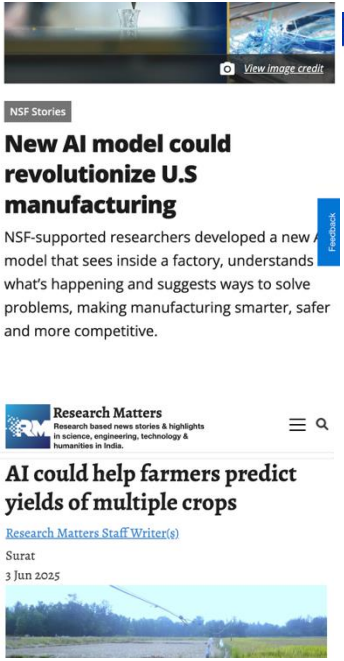
Kansas Is Built for Resilience → Uniquely Able to Surpass Digital Barriers

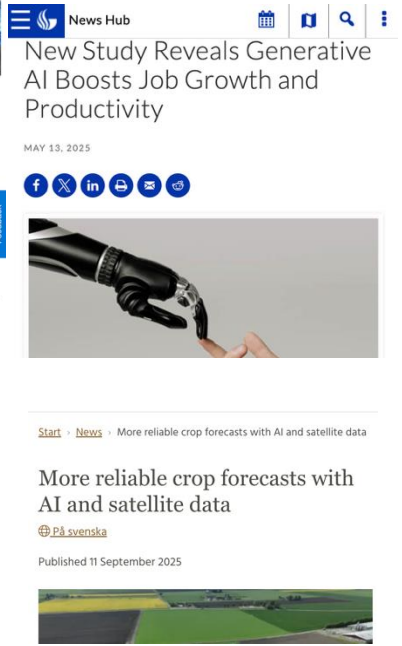
Source: Digital Commons Research (Elinor Ostrom / Governance Studies), AI and Blockchain Governance Reports, U.S. Infrastructure and Energy Studies, Innovation Geography Analyses

The Paradox: Promise and Peril of the Aurora

AI offers growth and competitiveness, yet Kansas must prepare for the long-term costs of inequity, fragility, and reliance.







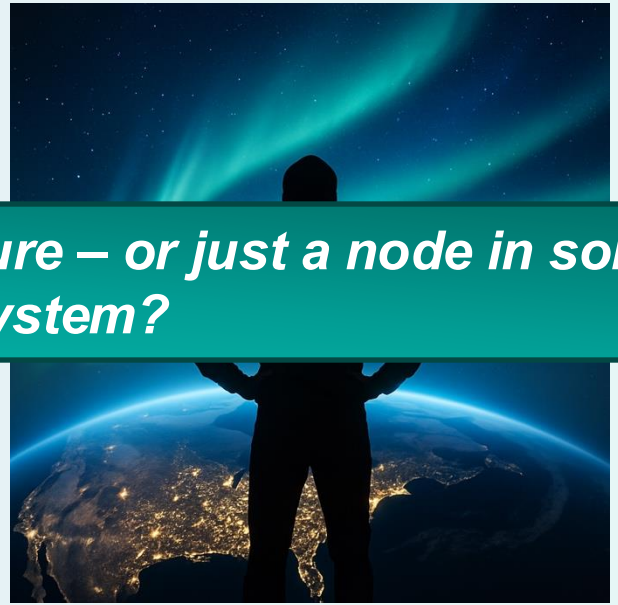
Equity Divide → benefits may concentrate in urban/elite clusters

Tech Dependence → reliance on external giants (Google, Meta, AWS) risks sovereignty

Fragile Supply Lines → global shocks (chips, rare earths, climate) could stall progress

Source: Business Insider, Deloitte, National Science Foundation (NSF), Research Matters, Crop Forecasting Reports (AI and Satellite Data)

Who Owns the Aurora?



Is Kansas the architect of its future – or just a node in someone else's system?

Google

Meta

Alphabet

HUAWEI

Source: Google, Meta, Apple, Alphabet, Huawei

Lessons From a Career in Global Clusters

Introducing a circular economy: New thinking with new managerial and policy implications

Authors Mark Esposito, Terence Tse, Khaled Soufani

COMPANIES IN THE CIRCULAR ECONOMY

Circular economy business models in developing economies: Lessons from India on reduce, recycle, and reuse paradigms

Sandeep Goyal[1] | Mark Esposito[2,3] | Amit Kapoor[1]

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Published Online: May 15, 2018 · pp 312-332 · https://doi.org/10.1504/WREMSD.2018.091688

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Is the Circular Economy a New Fast-Expanding Market?

Mark Esposito, Terence Tse, Khaled Soufani

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Journal of Management Development (2017) 36 (1): 58–70.

https://doi.org/10.1108/JMD-09-2014-0103 Article history

Literature review of emerging trends and future directions of e-commerce in global business landscape

Sandeep Goyal, Bruno S. Sergi and Mark Esposito

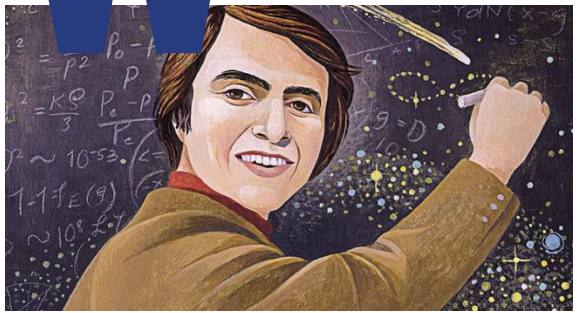
Published Online: March 27, 2019 · pp 226-255 · https://doi.org/10.1504/WREMSD.2019.098454

From Europe's industrial hubs to Asia's innovation corridors, clusters have been the lens to read resilience: Kansas is the next chapter.

Source: HAL Open Science, Journal of Management Development (Emerald Publishing), Thunderbird International Business Review, Arizona State University (Thunderbird School of Global Management)

From Cognitive Capital to Strategic Resilience

"The aurora borealis is a reminder that the most fleeting lights can reveal the greatest power."



Carl Sagan
Astronomer

THANKS!